

Carson Ball

Athens, GA 30606 • (678) 836-4977 • cmb42038@uga.edu
[linkedin.com/in/carsonmichaelball](https://www.linkedin.com/in/carsonmichaelball) • github.com/cmb42038

EDUCATION

University of Georgia

Bachelor of Science in Computer Systems Engineering | GPA: 3.42/4.0

Athens, GA

Expected May 2028

Relevant Coursework: Design of Digital Systems (Verilog/FPGA), Digital Logic Design, Electronics I, Circuit Analysis, Linear Systems, Systems Programming, Data Structures, Applied Linear Algebra

TECHNICAL SKILLS

Languages: C, C++, Verilog, Java, Python, MATLAB, Bash, Assembly (basic)

Embedded & Hardware: STM32, bare-metal firmware, Basys 3 FPGA, Vivado, op-amp/transistor circuits, oscilloscopes

Tools: Git, GitHub, Linux/Unix, GCC, GDB, Valgrind, POSIX threads, Make, Maven

PROJECT EXPERIENCE

IEEE Micromouse Robot

C/C++, STM32

Embedded Systems Competition Team, IEEE UGA Student Branch

September 2026 – Present

- Developing bare-metal C/C++ firmware (with some assembly) on an STM32 for an autonomous maze-solving robot.
- Implementing flood-fill and weighted Dijkstra path planning alongside PID motor control for fast, stable navigation.
- Integrating IR distance sensors, wheel encoders, and an IMU for wall detection, odometry, and heading correction.

FPGA Digital Design Labs

Verilog, Vivado

CSEE 4270 – Design of Digital Systems

August 2026 – Present

- Designed a binary-to-hexadecimal 7-segment LED decoder in Verilog and deployed it to a Basys 3 FPGA board.
- Wrote testbenches in Vivado to exhaustively simulate all input combinations and verify waveforms before synthesis.
- Edited Basys 3 constraint files to map module I/O to board switches and Pmod ports.

Systems Programming Projects

C/C++, Linux

CSCI 1730 – Systems Programming

January – May 2026

- Built a multithreaded POSIX threads program to search grids up to 3,567 x 3,567, comparing 1–3 thread runtimes.
- Wrote a file-comparison tool with low-level Unix system calls, analyzing the speed of byte-by-byte vs. bulk reads.
- Compressed DNA sequences to 2 bits per base with bitwise operators, cutting storage by about 75% versus ASCII.
- Managed dynamic memory with malloc/free, checking for leaks with Valgrind and debugging with GDB.

WORK EXPERIENCE

University of Georgia Libraries

Athens, GA

Senior Student Assistant

May 2025 – Present

- Promoted from Student Assistant to Senior Student Assistant based on reliability and job performance.
- Monitor library facilities and resolve patron concerns, escalating issues to supervisors when necessary.

Carrabba's Italian Grill

Athens, GA

Host

July 2026 – Present

- Coordinate seating and communicate with guests, servers, and management in a fast-paced team environment.

CAMPUS & COMMUNITY INVOLVEMENT

IEEE UGA Student Branch

Athens, GA

Member

August 2026 – Present

- Participate in technical meetings, workshops, and networking events with students and industry professionals.
- Selected for the branch's embedded systems competition team, one of several engineering project teams.

HONORS AND AWARDS

Zell Miller Scholarship, State of Georgia

August 2024 – Present

Eagle Scout, Boy Scouts of America

April 2021